

# Does Option Trading Convey Stock Price Information? \*

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## Abstract

Motivated by the growing literature on options as non-redundant assets, we investigate how option trading impacts underlying stock prices in the cross section controlling for stock market activities. We find both option trades and quotes are able to predict future stock price movement but stock orders submitted by stock market investors do not have predictive ability. The predictive relations do not reverse at longer horizons, suggesting that the price impact of option trading is due to revelation of private information rather than transitory price pressure as a result of option market makers' delta hedging activity in the stock market. Consistently, this informational benefit from option trading increases in concentration of informed traders, difficulty of short selling, and options market activeness. Trading strategies based on option trading signals yield significant abnormal returns that cannot be explained by common risk factors.

*JEL Classification:* G14, G12, G13.

*Keywords:* Options; order flow; information asymmetry; delta; price discovery.

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# 1. Introduction

The feedback effect of option trading on the stock market has been continuously arousing interests and awareness in the finance literature. Other than option listing effects<sup>1</sup>, the cross market informational linkage has drawn a large body of empirical research with the focus on which market contains key information about future stock prices. Although results from lead-lag analysis are inconclusive, there is some evidence showing that option volume contains stock price information. For example, Cao, Chen, and Griffin (2005) show that net option volume has significant predicting power of stock returns before merger and acquisition announcement and Pan and Poteshman (2006) find buyer-initiated put-call ratio predicts stock returns in the cross section.

Despite the empirical evidence of price impact from option trading, little is known on how this price impact on the underlying security is realized. There are at least two theories predicting option trading can affect the underlying stock price.<sup>2</sup> The first is market incompleteness due to information asymmetry. Black (1975) argue that the great leverage of option contracts can attract informed traders to the options market. Easley, O'Hara, and Srinivas (1998) develop a microstructure model and show that informed traders will trade both stock and options in a pooling equilibrium when relative option leverage and option market depth are high. They also argue that the multiple contracts in the options market allow informed traders to hide themselves. If informed traders actively trade options, net option volume taken by the non-market maker party can reveal information about future stock prices. Further more, if option market makers learn this information by observing order flows, they will update option quotes to protect themselves, thus making option quotes also informative about future stock prices. Information in this context is general advanced knowledge about the firm and does not have to be material information. Thus informed trading in this article is not indifferent with inside trading. The other possible way for options to affect underlying prices is through the price pressure of stock orders placed by option market makers. Delta hedging is now a standard procedure performed by option market makers to remove their risk exposure to underlying stock

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<sup>1</sup>Empirical studies investigating the impact of option listing on stock market focus on price reaction (Conrad (1989), Detemple and Jorion (1990), and Sorescu (2000)), relaxation of short sale constraint (Figlewski and Webb (1993), Danielsen and Sorescu (2001), Mayhew and Mihov (2005), and Battalio and Schultz (2006)), return volatility (Conrad (1989), Skinner (1989), Bollen (1998), and Mazouz (2004)), and market quality (Jennings and Starks (1986), Kumar, Sarin, and Shastri (1998), Mendenhall and Fehrs (1999), Mayhew and Mihov (2004), and Danielsen, Ness, and Warr (2007)). We apologize for being unable to cite a thorough literature on this subject.

<sup>2</sup>Here we focus on the conditional price impact of option trading, i.e. price sensitivity to option order flows and quotes. We do recognize that there can also be a level effect on the underlying security price when derivative securities are introduced. Readers interested in asset pricing implication of option introduction can refer to Ross (1976), Hakansson (1982), and Figlewski and Webb (1993).

price movement. As a result, option transactions will lead to consequent stock transactions. Chordia and Subrahmanyam (2004) show that it is optimal to split exogenously given orders over several periods for large liquidity traders such as financial institutions in order to minimize the overall price impact and the autocorrelated order imbalance can therefore predict the movement of stock prices. The model can be applied to the options market. Similarly, positive autocorrelation in option order imbalance will be positively correlated with option price movement. As a result of delta hedging, the option order imbalance can also have predictive ability about future stock price movement although no additional private information is revealed.

Although both theories suggest a real impact on the underlying price from option trading, they disagree substantially on an important question, whether the predictive ability of option order flows is impounded by stock order flows. The information theory regards option order flows as an independent source of information about future stock prices, suggesting the same information is not necessarily available in the stock market. Although option market makers will pass the information to the stock market through delta hedging, other market participants are not able to observe this information transmission directly in the stock market because option market makers anonymously trade stocks. Investors must exert extra effort to learn the information in the options market. The price pressure theory, however, implies that the predictive ability of option order flows may disappear once stock order flows are controlled for. Since option market makers do not reveal their identity to the general public in the stock market, there is no reason to assume that the price pressure of option market makers' orders will be different from that of another stock market investor's orders. It is the overall stock order imbalance from all stock market participants that determines the price pressure on the stock. Therefore, investors do not need to watch option order flows but only stock order flows. The two theories also differ on price impact at longer horizons. The information theory predicts that the price impact is permanent while the price pressure theory predicts a transitory impact that will reverse to fundamentals at longer horizons.

In this paper, we investigate the mechanisms of how option trading affects the stock price and the potential information transmission channels. We decompose the overall stock order imbalance into a delta hedging component from order imbalance in the options market and a pure stock market component. The decomposition allows us to separate the sources of order flow information and price pressure from different markets. We then investigate whether option order flows contain additional price information controlling for stock order flows and other microstructure impacts. Option quotes may also provide additional information

because quotes reflect option market makers' belief conditioning on option order flows they observe. If there exists some information segmentation preventing rational agents from observing option order flows or if this information is too costly to acquire and interpret, non-option market makers will not be able to update their belief using the full information set and the spot price will be inefficient. We test this hypothesis by examining pricing effect of the difference between the option implied stock price and the spot price in the cross section.

This paper contributes to the finance literature in several ways. First, we find that pure stock market order flows have surprisingly little predictive ability in the sample of optionable stocks but option order flows have strong and robust predictive ability of the next day's stock price movement. The predictive relations do not reverse at longer horizons, indicating that the price impact from option trading is due to information asymmetry among investors rather than transitory price pressure on the stock market through delta hedging, thus providing new evidence on where informed traders trade. On the other hand, stock order flows can significantly predict future returns for stocks without options during the same period. It seems when options are available, informed traders migrate from the stock market to the options market. Many previous studies also link option transactions to stock prices, but do not control for stock transactions, e.g. Easley, O'Hara, and Srinivas (1998) and Pan and Poteshman (2006). In the controlled setup, Chan, Chung, and Fong (2002) systematically examine the information content in net stock and option volumes but fail to establish a link between net option volume and stock prices. Cao, Chen, and Griffin (2005) also conclude that the options market is only informative right before "extreme events" such as merger announcement but not during normal periods. As good as to our knowledge, we are the first to document significant pricing effect of option order flows controlling for stock order flows. The difference in findings in this paper with previous studies is mainly due to two reasons. First, our data covers a larger proportion of optionable stocks in the US in a more recent sample from 2008 to 2010. The options market has experienced tremendous consolidation and growth since 2000 in response to SEC's requirement of establishing intermarket linkage. Aside from increased competition, the prevailing electronic trading and automated quote updating also significantly increase market liquidity, hence reducing transaction costs for informed traders. It is possible that the level of informed trading in the options market is higher in our sample period than periods in previous studies.

<sup>3</sup> Second, more importantly, the option order imbalance measure in this study is derived from delta risk exposure of all option contracts on the same underlying security while previous studies usually rely on

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<sup>3</sup>Ideally we would like to extend our analysis to early periods to test this conjecture. We are not able to do so due to data availability.

only one pair of options or simply aggregate total option volume. Measuring information in option trades is not straight forward as there can be multiple options underlying the same stock with different strikes and maturities. Holowczak, Hu, and Wu (2011) show the optimal information aggregation method must consider option risk exposure. Therefore, in this paper we aggregate the net delta position instead of net option volume to more accurately measure the price information in option order flows.

Additionally, this paper confirms the informational role of option quotes in price discovery. If option order flows contain price information, option market makers can learn the information and adjust option quotes to reflect this information. Similarly, stock market makers can learn the price information in stock order flows and the spot price reflects such information. Unlike most studies focusing on the time series relation between option implied and actual prices, we investigate the cross sectional informational linkage to see if order flow information in one market is learnt on the other market. We find the prices are generally consistent on the two markets. However, the price discrepancy between option implied and actual prices has significant predictive ability of future stock returns up to three weeks ahead, suggesting stock market participants do not fully learn the information in option order flows. Therefore, other than option order flows, option quotes is also an important channel for information transmission from the options market to the stock market. The data set we use has an advantage over previous studies, i.e. after 2007, both options and stock markets close at the same time, 4 pm EST while the options market used to close after the stock market. There is no time synchronicity issue to use the close prices in our study.

This paper also contributes to a strand of literature that investigates the interaction of market frictions and firm characteristics. We examine the predictive ability of option trades and quotes conditioning on firm size and institutional ownership. Small firms are more likely to be subject to information asymmetry as they have less analyst coverage. Therefore, it is expected that trading process should convey more information for small stocks. However, small stocks are more likely to suffer from liquidity shocks, which can also lead to order imbalance and price discrepancy but do not need to be informational. The overall impact of firm size on the informativeness of option trading is thus unclear. Our empirical results show that the information content in option trading process is the greatest for small firms, consistent with the hypothesis that small firms have higher levels of information asymmetry. Firms with diversified ownership are more likely to be subject to short sale constraint because it is more difficult to borrow a large amount of stocks to meet short interests on such stocks. As a result, informed traders may have to place more orders in the options market as an alternative to short as discussed by Figlewski and Webb (1993). Option trading makes the stock price

more efficient by relaxing the constraint on short selling. We expect the informational benefit is stronger for stocks with more short sale constraint proxied by institutional ownership. The results support this hypothesis as both option trades and quotes have greater predictive ability for firms with diversified ownership.

We also show that the predictive ability of option order imbalance is not due to illiquidity effect. Option order flows become more informative when the options market is active, consistent with the theoretical prediction in Easley, O'Hara, and Srinivas (1998) that informed traders prefer to trade in deep markets. On the contrary, stock order flows have more information when stock volume is low. We find the information in the options market implies trading profitability. Strategies based on option order imbalance and price discrepancy yield significant returns that cannot be explained by market risk, size, book-to-market ratio, and return momentum.

The rest of the paper is organized as follows. In section 2, we review the literature on conditional role of options in transmitting private information. In section 3, we propose a measure of option order imbalance and develop the hypotheses for empirical tests. In section 4, we describe our data in this study. In section 5, we report the empirical results of the hypothesis tests. In section 6, we further exploit the predictive ability of option trades and quotes in additional analysis. And section 7 concludes.

## **2. Motivation**

A rich body of the finance literature has studied the informational linkage across the options market and the stock market. The use of options market as an alternative trading venue by informed traders has been theoretically examined by Black (1975), Biais and Hillion (1994), and Easley, O'Hara, and Srinivas (1998). The main conclusion from these papers is that informed trading can take place in the options market depending on market liquidity and leverage achievable. A group of researchers find option volume is able to predict future stock returns as empirical evidence to support the theory, e.g. Easley, O'Hara, and Srinivas (1998), Poteshman (2006), Pan and Poteshman (2006), Holowczak, Hu, and Wu (2011) and Dong and Sinha (2011). Although the empirical results are quite robust, these works still face the challenge from an alternative explanation that the price impact is simply a mechanical price pressure as a result of delta hedging.

Option market makers trade stocks to manage their risk exposure to underlying stock price movement

and the optimal hedge ratio is captured by the option price sensitivity to the underlying price, delta. Delta hedging will transform option orders into stock orders, causing a transitory price pressure in the stock market. For example, when there is large buy pressure on call options or large sell pressure on put options, option market makers need to buy stocks to hedge, causing buy pressure on the stock market subsequently. The reverse is true when there is sell pressure on call options or buy pressure on put options. The price pressure from delta hedging demand, however, faces the competition from orders submitted by the rest of the market, which we characterize as pure stock market orders. Option market makers trade anonymously in the stock market so their delta hedging orders are mixed with pure stock market orders. It is thus the overall stock order flows not delta hedging orders alone that lead to the price pressure on the stock. When these two are positively correlated or when delta hedging orders dominate pure stock market orders, the stock price pressure may appear to come from delta hedging, hence option order flows.

There is a growing literature investigating the autocorrelation of stock order flows and the consequent pricing effect. A number of papers study order placing techniques in the stock market and conclude that the optimal trading algorithm is to split the exogenously given order and trade in several periods, for example, Bertsimas and Lo (1998), Vayanos (2001), and Chordia and Subrahmanyam (2004). Although the analyzed security is stock in all these models, the main results can be applied to derivative markets because the key noise in the underlying price dynamic comes from trade occurrence. Empirically, Chordia and Subrahmanyam (2004) show that stock order flows are positively autocorrelated at daily frequency and are able to positively predict the next day's returns in the cross section. They also show that the price impact is transitory as the predictive correlation reverses sign after one day. Heston, Korajczyk, and Sadka (2010) examine the trading patterns of half-hour intervals and find strong autocorrelation of order imbalance in the same thirty-minute trading periods over multiple days. Combining these findings with the fact that option market makers' delta hedging activity transforms option order flows into stock order flows, it is thus unclear whether the predictive ability of option orders represents *additional* information in the stock market or if such predictive ability even exists on top of stock order flows. If all price impact in option order flows is only realized through stock order flows, then the options market plays a trivial role in transmitting information.

<sup>4</sup> Therefore, it is important to control for stock order flows when investigating the information content in option transactions.

Of course we are not the first to examine the information content of option order flows in control of

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<sup>4</sup>In this case, the options market functions by allocating risk only but can still be non-redundant.

stock order flows. Chan, Chung, and Fong (2002) investigate the informational role of option and stock volume together with option and stock quote revisions in a sample of 14 most actively traded option tickers in the first quarter of 1995. They find no pricing effect of net option volume as evidence against informed trading in the options market. They show net stock volume and both option and stock quote revisions have predictive ability. Cao, Chen, and Griffin (2005) study the impact of option and stock order imbalance for 78 takeover targets between 1986 and 1994. Their results suggest option volume imbalance becomes significantly informative about future stock prices right before merger announcements but remains silent during normal periods. The limited sample size and crucial measure of option order flow information in these studies motivate us to investigate the question in a larger sample using a refined option information measure. The market structure at the time of these studies is also very different from the present (Battalio, Hatch, and Jennings (2004)). We suspect the same results may not hold any more.

Other than option volume, option prices may also contain information about the underlying security. In the model of Easley, O'Hara, and Srinivas (1998), risk neutral market makers provide quotes based on updated belief and the quotes reflect private information. Cao (1999) predicts options increase the incentive of information collection and improve price efficiency if private information is costly to acquire. Biais and Hillion (1994) show that the existence of options market can mitigate the market breakdown problem if uninformed traders perceive the benefit of better risk sharing opportunities to be greater than the potential loss of trading with informed traders. However, the impact on information efficiency is unclear because information extraction is difficult in the options market although more private information can be revealed. Huang and Wang (1997) also reach the same conclusion that information efficiency may be reduced if hedging trades generate too much noise in the options market.<sup>5</sup> The information efficiency gain from option trading can be more complicated if agents are not able to simultaneously update belief on trade occurrence. Most theoretical models assume order flow information is public for rational agents to solve the equilibrium price. However, this information is costly to acquire and interpret in reality, especially when trading is active. As a result, market makers have an advantage over other investors of learning price information in order flows *on that market*. If such information segmentation exists, there may be discrepancy between option implied and actual prices. Therefore, option quotes may have additional information that is not in the underlying market if the cost of price information in option order flows is too high for non-option market makers. The reverse argument can also be true if the cost of stock order flow information is too high for

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<sup>5</sup>Empirically, several studies find the price reaction to earnings announcement become smaller and the adjustment speed increases after options are traded, suggesting the market becomes more informationally efficient (e.g. Jennings and Starks (1986) and Skinner (1990)) while others find the opposite results (e.g. Mendenhall and Fehrs (1999))

non-stock market makers. Early empirical lead-lag analysis of the question generates mixed results. For example, Manaster and Rendleman (1982) and Bhattacharya (1987) find option implied price leads the spot price by one day. But their results are clouded by the fact that the options market closed after the stock market during their sample period and information arriving after stock market close can only be reflected in the options market. Stephan and Whaley (1990) find the reverse relationship using intraday transaction data. Chan, Chung, and Johnson (1993) report no significant relation between stock return and the price implied from option mid-quotes. Using the approach of information share developed by Hasbrouck (1995), Chakravarty, Gulen, and Mayhew (2004) and Holowczak, Simaan, and Wu (2006) document option quotes also contribute to price discovery although the majority contribution comes from the stock market. More recently, Muravyev, Pearson, and Broussard (2011) find that it is option quotes but not stock quotes that adjust to eliminate the arbitrage opportunities and conclude that there is no role for option quotes in price discovery. Time series studies can be confined to a small sample of options. We take a different approach by investigating the cross sectional relation between option quotes and future price movement.

### **3. Methodology**

In this section, we describe the empirical test we design to uncover how option trading impacts the underlying stock prices. We will begin with how to measure price information in options and stock order flows and quotes. After that, we will introduce the main hypotheses and empirical testing methods.

#### **3.1. Option Order Imbalance**

A remarkable feature of the options market is that there can be hundreds of option contracts underlying the same security. For example, on January 2 2009, there are 478 option contracts underlying the common stock of Apple Inc., spanning 61 strike prices between \$12.5 and \$400<sup>6</sup> and 6 maturity dates in 15 days to 750 days. As each contract has its unique combination of put/call type, strike price and maturity, the trading process of each option contract does not necessarily contain underlying price information in the same way. Measuring order imbalance in the options market is challenging. In a recent study, Holowczak, Hu, and Wu (2011) show the risk exposure to the underlying stock price (delta) is an important consideration in aggregating stock price information in option transactions. Following their intuition, we propose a standardized

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<sup>6</sup>The close price is \$90.75 on that day.

measure to capture the order imbalance in the options market:

$$OOI_{it} = \frac{\sum_{j=1}^N 100Dir_{itj} \cdot \delta_{itj} \cdot size_{itj}}{Num\_shares\_outstanding}. \quad (1)$$

Option order imbalance,  $OOI_{it}$ , is measured for stock  $i$  on day  $t$ .  $Dir_{itj}$  is a dummy variable equal to 1 when the trade is initiated by the buyer and -1 when the  $j^{th}$  trade is initiated by the seller according to certain trade signing algorithm.  $\delta_{itj}$  is the option price sensitivity to the underlying stock price of trade  $j$  and  $size_{itj}$  denotes its trade size in option lots (100 shares of the underlying stock). The numerator is thus the net delta position sought by non-market makers in the options market. Option market makers will take exactly the opposite delta position. In order to hedge the delta risk, market makers will then have to transact the same amount of stocks in the stock market. Although it is unknown when option market makers will perform delta hedging after they receive option orders, we can still infer the stock demand of option market makers on a daily basis because option market makers tend to go home “flat”, i.e. they normally keep delta neutral positions overnight. Note that option market makers do not necessarily wait until the end of the day to hedge when the net delta position is known. It is difficult to track all stock orders submitted by option market makers and the total transaction volume can be much higher than the numerator in equation 1. Nonetheless, one can learn option market makers’ daily net demand of stock in number of shares if the hedge ratio does not fluctuate too much intra-day. It is difficult to measure the net number of trades transacted by option market makers because they can split or aggregate hedging transactions. So we measure option order imbalance in terms of number of shares but not number of trades.

Order flows are more likely to be imbalanced when trading is inactive and the imbalance may reflect mere noise rather than information. To address the liquidity concern, we scale the net delta volume with the number of shares outstanding for cross sectional standardization instead of shrinking our sample size by choosing an arbitrary volume cutoff. Our option order imbalance measures the directional trading intention in the options market relative to the total shares available in the stock market.<sup>7</sup>

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<sup>7</sup>We have also experimented with alternative scalars such as total delta volume and total stock volume for robustness check. Our main results still hold qualitatively but these alternative scalars are more likely to generate outliers during inactive trading periods. Hence we do not report these results in the paper.

### 3.2. Stock Order Imbalance

Defining option order imbalance in terms of delta imbalance has another advantage as now we can calculate the stock order imbalance in the rest of the market by removing delta imbalance from total stock order imbalance. Subsequently, the pure stock order imbalance is defined as

$$SOI_{it} = TOI_{it} - OOI_{it} = \frac{\sum_{k=1}^N Dir_{itk} \cdot size_{itk}}{Num\_shares\_outstanding} - OOI_{it}, \quad (2)$$

where  $TOI_{it}$  is total stock order imbalance in terms of trade volume, and  $Dir_{itk}$  and  $size_{itk}$  are direction dummy variable and size for the  $k^h$  trade of firm  $i$  on day  $t$  in the stock market, respectively. We also scale this order imbalance with total number of shares outstanding to keep it consistent with option order imbalance. After subtracting option order imbalance, the remaining is simply the stock order imbalance generated by stock market investors other than option market makers. Note that although measuring option order imbalance is also feasible at shorter intervals than one day, the decomposition of total stock order imbalance is less likely to be valid at higher frequencies because option market makers may not perform full delta hedging within the same observation period.

### 3.3. Price Discrepancy in Quotes

Our tests also involve option quotes and we need to measure the stock price information from the quoted call and put prices. We use a model-free approach to construct an option implied stock price from the put-call parity as:

$$S_{implied} = C - P + Ke^{-rt}, \quad (3)$$

where  $C$  and  $P$  are prices of call and put options with the same strike  $K$  and time to maturity  $t$  and  $r$  is the risk-free rate. We omit firm and time subscriptions in equation 3. Since exchange listed equity options are American style and may not be subject to the put-call parity, we use the pair of one-month closest-to-the-money options to construct implied stock price for each stock because these options are the least likely to be early exercised. Then the price discrepancy on the two markets is defined as

$$PD = \frac{S_{implied} - SpotPrice}{SpotPrice}. \quad (4)$$

We scale the price difference using the spot price for cross sectional standardization.

### 3.4. Main Test

Our goal in this study is to understand whether option trading provides additional information about the underlying security price on top of stock order flows and spot prices. To identify the price discovery contribution of security trading in the two markets, we estimate the following equation in the cross section:

$$Ret_{i,t+k} = \alpha^k + \beta_1^k SOI_{i,t} + \beta_2^k OOI_{i,t} + \beta_3^k PD_{i,t} + \theta^k X_{i,t} + \varepsilon_{i,t+1}, \quad (5)$$

where  $Ret_{i,t+k}$  is the four factor risk-adjusted return of stock  $i$  on day  $t+k$ , and  $X_t$  is a set of control variables, including closing bid-ask spread of the stock and stock turnover ratio on day  $t$  to control for trading volume studied by Chordia and Swaminathan (2000) and Gervais, Kaniel, and Mingelgrin (2001), and cumulative stock return from day  $t-5$  to  $t-1$  to control for short term reversal effect documented by Lo and MacKinlay (1990). In theory, the private information in order flows is about specific firms rather than the whole market and should have more predictive ability about idiosyncratic returns. We follow the standard method in the literature and use a four-factor model to remove from raw returns the compensation for systematic risk to get our dependent variable.

*Hypothesis 1: Stock order imbalance contains information about future stock price movement, i.e. at least for some  $k$ ,  $\beta_1^k \neq 0$  and the signs do not reverse.*

The first hypothesis focuses on predictive ability of stock order imbalance. Similar to the price impact from option trading, there can be two different ways that stock trading impact future price movement. Informed trading can happen in the stock market, making stock order flows informative. The transitory price pressure and positively autocorrelated stock order flows can also lead to a positive correlation between stock order imbalance and future returns. If the relation is due to informed trading, then the price impact is permanent after new information is embedded into stock prices. If the relation is due to transitory price pressure, a reversal relation is expected after certain period of time. We will then regress returns from further future periods on the same independent variables to disentangle the sources of price impact.

*Hypothesis 2: Option order imbalance contains additional information about future stock price movement that is not in spot price and stock order flows, i.e. at least for some  $k$ ,  $\beta_2^k \neq 0$  and the signs do not reverse.*

Either informed trading in the options market or price pressure induced by option market makers' delta

hedging will generate a price impact from option transactions. Informed traders will purchase/sell delta on good/bad private information, causing a positive/negative option order imbalance. Option market makers' delta hedging demand also has a positive correlation with stock price pressure. The net option order imbalance is thus expected to be positively related to future stock returns. Under price pressure theory, the price impact of option transactions goes through stock transactions and a positive predictive relation is also expected. However, the sign of  $\beta_2^k$  will reverse in longer horizons.

*Hypothesis 3: Price discrepancy between option implied price and spot price also contains price information, i.e. at least for some  $k$ ,  $\beta_3^k \neq 0$  and the signs do not reverse.*

We investigate another possible channel of information transmission across markets through option quotes. If option order flows contain information about future stock prices, option market maker may learn the information. If other investors face higher costs to acquire this information, option quotes can have additional information controlling for the spot price and order flows.

### **3.5. Level of Information Asymmetry**

We are interested in the heterogeneity of price sensitivity to option trades and quotes across firms. If the price impact of option trading comes from informed trading, we expect the predictive ability of option order imbalance and price discrepancy to be stronger for firms that are more likely to have information asymmetry. Small firms usually have less analyst coverage and are more likely to attract informed traders. Therefore, we have the following hypothesis.

*Hypothesis 4: If option trading reveals private information, the predictive ability of option order imbalance and price discrepancy has a negative correlation with firm size, a variable to proxy for concentration of informed traders.*

To test this hypothesis, we reestimate equation 5 in subsamples of three size groups based on NYSE breakpoints: small ( $< 30\%$ ), medium ( $30 \sim 70\%$ ), and large ( $\geq 70\%$ ). Comparing the magnitude and significance of estimated coefficients across groups may not be the best way to test this hypothesis because the variables can have very different statistical features in subsample groups. Alternatively, we compare the economic significance of estimated coefficients by calculating contribution to idiosyncratic returns from one standard deviation change of independent variables.

### 3.6. Short Sale Constraint

Other than level of information asymmetry, the easiness of short selling will also have an impact on the price impact of option trading. It is known that options are used as a device to get around the regulation on short selling (Figlewski and Webb (1993)). For stocks that are more difficult to short, option trading can have larger informational benefit as investors with negative views may be forced to trade options only. Institutional ownership as a proxy for market supply of short interest is known to be negatively related with the difficulty of short selling (Asquith, Pathak, and Ritter (2005)). Thus we have the following hypothesis.

*Hypothesis 5: The informational benefit of option trading is greater for firms with diversified ownership because it is more difficult to borrow shares of these firms from institutional investors to sell short.*

To test this hypothesis, we reestimate equation 5 in subsamples of three institutional ownership groups: low ( $< 30\%$ ), medium ( $30 \sim 70\%$ ), and high ( $\geq 70\%$ ) and compare the economic significance of estimated coefficients in subsample groups.

### 3.7. Information v.s. Illiquidity

Although our cross sectional standardization of independent variables helps mitigate the outlier problem caused by liquidity shocks, there can still be concerns regarding interpretation of our results because order imbalance may reflect market illiquidity. Illiquidity has been known to have a pricing effect in the stock market (Amihud (2002)). In a recent study, Christoffersen, Goyenko, Jacobs, and Karoui (2011) find the options market also has an illiquidity premium. Liquidity can be a big concern in options market microstructure studies because many option tickers may not be actively traded. That is why previous studies usually analyze only a group of active option tickers. We address the liquidity concern by investigating the pricing effect of order flows and quotes conditioning on trading volume as a proxy for market liquidity. If the pricing effect reflects market illiquidity, we expect the impact is weaker when trading volume is high as the market becomes more liquid. On the other hand, if the predictive relation between order flows and future stock returns is a result of informed trading, it is expected that the information content will increase when market is more active because informed traders can better disguise among uninformed traders and lower their transaction costs.

*Hypothesis 6: For both stock and options, if the pricing effect of order imbalance is a result of illiquidity,*

*it should decrease when total trade volume increases in that market; alternatively, if the pricing effect of order imbalance is a result of informed trading, it should increase when total trade volume increases.*

To test this hypothesis, we divide the full sample into three option volume groups and three stock volume groups separately and reestimate equation 5 in subsamples.

## **4. Data**

The joint investigation of trades and quotes on both stock and options markets requires merging several databases. We describe the details of data sources, sample selection, and variable construction in this section.

### **4.1. Option Market Activity**

We obtain option transactions data between April 2008 and August 2010 from Trade Alert LLC, a specialized option market data vendor. The data includes all trade messages recorded by the Option Pricing Reporting Authority (OPRA), a national information center as a result of SEC's requirement of building a consolidated options market across US options exchanges. Trade Alert matches option transaction data with the spot market data and computes option implied volatility for each transaction from a binomial tree in real time. Trade direction is classified based on the quote rule. If the trade price is above the last effective mid quote price, it is classified as buyer-initiated. If the trade price is below the last effective mid quote price, it is classified as seller-initiated. The comprehensive trade data allows a cross sectional study in a relatively long time period.

In table 1 we present the main statistics for the whole options market as well as for four transaction type groups, i.e. buy call, sell call, buy put, and sell put. Since the quote rule is not able to classify trade directions when the trade price falls exactly on the mid quote or when there is no valid quote at all, there is another group of option trades in the sample which cannot be classified into any of the four transaction type groups. We do not report statistics of this group because these trades are not of interest in the study.<sup>8</sup> There are 2,605 underlying securities on an average trading day that have at least one trade record during the sample period with the maximum at 2,933 and the minimum at 2,225. Call options are traded more

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<sup>8</sup>Omitting this group, however, may cause the full sample statistics to be inconsistent with the sum of statistics across the four transaction type groups.

often than put options as the average daily number of firms with call transactions exceeds the average daily number of firms with put transactions in both buy and sell categories in panel A. Trade statistics are reported in panel B. The average trade size is about 23 lots in the whole sample. Call option trades are on average smaller than put option trades. However, the average daily number of call options is much greater than the average daily number of put options. As a result, the average daily volume of call options also exceeds the average daily volume of put options by about 1 million lots. The total premium in the options market reaches 3 billion dollars a day on average. Lakonishok, Lee, Pearson, and Poteshman (2006) find non-market maker participants take a net short position in option open interest but a net long position in daily option volume between 1990 and 2001. Our data confirm their finding as panel B shows for both call and put options, the daily average buy trades exceed the daily average sell trades in both volume and premium. Panel C presents the distribution of trade volume across option moneyness and time to expiration. Across moneyness, out-of-the-money (OTM) options are the most heavily traded, accounting for more than 50% of the entire market volume. At-the-money (ATM) options account for 32.81% of the total volume and in-the-money (ITM) options has the smallest share of only 16.82%. The same pattern can be found in the four transaction type groups. Also across three moneyness regions, non-market maker participants take a net long position except for ITM puts. Across maturities, about 45.69% of the transactions are on options expiring in 30 calendar days. Options expiring in 31 to 60 days account for 25.1% of the total volume and options with maturity longer than two months account for the rest 29.21%. The same pattern exists in the four transaction type groups and buy volume always exceeds sell volume.

At the beginning of our sample period, there were seven options exchanges in the OPRA plan, American Stock Exchange (AMEX), New York Stock Exchange ARCA (NYSE), Boston Options Exchange (BOX), Chicago Board Options Exchange (CBOE), International Securities Exchange (ISE), NASDAQ, and Philadelphia Stock Exchange (PHLX). Another participant exchange joined the OPRA during our sample period as BATS launched its options trading platform on February 26, 2010. The last panel of table 1 presents the volume share by exchanges. ISE and CBOE lead the market share as each of them attracts more than one quarter of the total volume. PHLX is also very competitive by attracting about 19% of the total volume, followed by NYSE of 12.76% and AMEX of 8.37%. BOX and NASDAQ are relatively small markets as neither has more than 4% of the total volume. BATS does not have much market share during our sample period because it was newly established. The Herfindahl index of market share is 0.203, indicating the options market in the US is moderately concentrated during our sample period.

To be included in the sample, an observation must have at least one valid option transaction on that day. As a result, we have an unbalanced panel of firm-days. When computing option order imbalance, we exclude the following trades: (i) off-hour trades; (ii) trades at the market open (the first fifteen minutes) and market close (the last five minutes); (iii) trades that are reported late to the OPRA; (iv) data errors such as 0 strike price or 0 trade price; (v) trades that cannot be classified as either buyer- or seller-initiated. We filter out (i), (ii), and (iii) because trade classification is less reliable for these trades. Since we focus on firm level information asymmetry, we exclude all index and ETF options. We also exclude stock-split days and dividend days for their complex implication on option pricing and trading. On average, there are 2,275 firms a day in the final sample and each firm has an average trading volume of 161 trades and 3,118 lots a day.

## 4.2. Other Data Sources

To construct the option implied stock price from option quotes, we use the midpoint of closing quotes in Option Metrics. To make sure the American option feature of equity options does not distort the put-call parity, we only choose a pair of call and put options with the strike closest to the spot price and maturity closest to one month. During our sample period, both markets close at 4 pm EST so there is no synchronicity issue.

We obtain stock transaction data from NYSE Trade and Quote (TAQ) database. Our stock order imbalance also needs the information on trade direction. We follow Lee and Ready (1991) in signing trades. After applying the quote rule, if the trade is still unclassified, we compare the trade price to the last different trade price. If the current trade price is higher/lower than the last different price, it is classified as buyer/seller-initiated. Unlike Lee and Ready (1991), however, we collapse trades in the same second into one trade record weighted by dollar volume and use the most updated National Best Bid and Offer (NBBO) one second before the trade time ( $t - 1$ ) to perform trade signing. In addition to excluding canceled trades and data errors, we also exclude trades within fifteen minutes after market open and five minutes before market close to increase the accuracy of trade signing and match the observation period of option order imbalance.

Finally, stock return data is collected from CRSP and institutional ownership data is extracted from 13-F filings. Fama-French factor returns and NYSE size breakpoints are downloaded from French's website.<sup>9</sup>

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<sup>9</sup>We thank Kenneth French for making the data available at <http://mba.tuck.dartmouth.edu/pages/faculty/ken.french>

### 4.3. Statistics of Main Variables

We construct stock order imbalance, option order imbalance, and price discrepancy as discussed earlier. When calculating option delta, we assume 0 interest rate and dividend rate for simplicity. Table 2 presents the main statistics of these variables. The two order imbalance variables are constructed using transaction data from 9:45 am to 3:55 pm every day and the price discrepancy is measured using market close prices at 4 pm. Panel A gives the descriptive statistics for the full sample. For reporting easiness, we scale return, stock order imbalance, and option order imbalance to basis points and report price discrepancy in percentage. The first thing to notice is that the average order imbalance is close to 0 (less than 1 bp) on both stock and options markets, indicating that both markets are generally well balanced. Stock order imbalance has both larger mean and variance than option order imbalance but option order imbalance has fatter tails than stock order imbalance. Average daily minimum and maximum stock order imbalance are -5% and 4%, respectively. However, firm level stock order imbalance can exceed -48.5% and 57.4% in extreme cases. Comparatively, option order imbalance has smaller magnitude. Average maximum and minimum option order imbalance are 1.3% and -1.4%, respectively while the maximum and minimum firm level option order imbalance can reach 18.7% and -27.2%, respectively. The distributions of stock and option order imbalances are plotted in figure 1 after deleting the top and bottom 0.5% observations.

In order to test if option order imbalance dominates stock order imbalance, we compute the fraction of absolute values of these two variables. The ratio will explode when stock order imbalance is close to zero. We winsorize this variable at top and bottom 0.5% level and report the statistics in panel A of table 2. On average, option order imbalance is about 45% of stock order imbalance but the standard deviation is high. The mean may overstate the relative magnitude of option order imbalance if winsorization does not fully address the outlier issue. Actually the median ratio of the two order imbalances is only 6.74%. Only 42.11% observations in our sample have option order imbalance greater than ten percent of stock order imbalance in absolute value. The proportion drops to 15.47% when the benchmark is raised to half of stock order imbalance and only 8.34% observations have option order imbalance greater than stock order imbalance in absolute value. We also plot the distribution of this variable in figure 1. It is clear that for majority of observations, option order imbalance is less than 5% of stock order imbalance. Therefore, it is unlikely that option order imbalance can dominate stock order imbalance in determining the direction of overall imbalance in the stock market.

The average price discrepancy is about -0.16% of the spot price, suggesting the option implied price

is generally consistent with the spot price. However, there are firms with almost zero option implied price as the minimum price discrepancy is below -99%. On the other hand, the maximum option implied price can be more than 17.6 times the spot price. The time series average of daily minimum and maximum price discrepancy are -67.48% and 126.17%, respectively.

[Figure 1 about here.]

We also report statistics of trading volume on the two markets. The option volume unit, one lot, consists of one hundred shares of the underlying stock. We report option trade volume as the natural logarithm of number of option lots traded. To be consistent, we report stock volume as the natural logarithm of hundreds of stock shares traded. It is not surprising that the stock market volume is larger than the options market volume. However, option volume has a larger cross sectional standard deviation than stock volume. This is due to the fact that many option tickers are inactively traded. After taking logs, the trade volume distributions also become smooth as the skewness and excess kurtosis of both stock and option volumes are small.

In panel B of table 2, we report the time series average of daily cross sectional correlations between main variables. The surprising result is that the correlation of stock order imbalance and option order imbalance is large and negative (-0.173), suggesting investors on the two markets initiate trades at the opposite directions on average. 50.07% of our observations have the two order imbalance variables with different signs. A natural explanation of the finding is that investors use options to hedge their newly established stock positions, e.g. covered calls and protected puts. Such structured trades are often out of the motivation of private information and the price pressure should cancel out in the stock market. Thus the price impact is expected to be neutral. Although structured trades are not supposed to bias our results in any direction, we are concerned with the unknown motivation behind them without more detailed information. In unreported robustness checks, we do our best to eliminate the impact of structured trades by excluding all option transactions flagged by OPRA as part of structured trades involving stocks or other options. Our results are not affected by this treatment. The fact that option order imbalance is negatively related with stock order imbalance and does not dominate the latter implies the price pressure caused by delta hedging can be different from the overall market price pressure.

The negative correlation (-0.006) between stock order imbalance and price discrepancy is as expected because holding everything else the same, positive stock order imbalance will push the stock price to a higher level, thus making our price discrepancy measure more negative. What's surprising is price discrepancy has a

weak but negative correlation (-0.001) with option order imbalance because positive option order imbalance should push up the implied stock price in the options market, making our price discrepancy measure more positive. The insignificant correlation between option order imbalance and price discrepancy suggests that option order imbalance does not necessarily lead to price discrepancy and these two variables can represent different price impact channels from option trading. The correlation between option and stock volumes is large (0.704).

The cross sectional average of daily autocorrelations are presented for each variable up to five lags in panel C. Short term reversal effect is found for stock returns up to five days as the autocorrelations are all negative. Stock order imbalance has highly positive autocorrelations and the first-lag autocorrelation reaches 0.094. The autocorrelation of stock order imbalance decreases gradually and the fifth-lag remains 0.024. Option order imbalance is also positively autocorrelated but the autocorrelations are much smaller than stock order imbalance. Even the first-lag autocorrelation is only 0.033. The positive autocorrelation of order imbalance on the two markets is consistent with the optimal trading strategy documented in Bertsimas and Lo (1998) and Chordia and Subrahmanyam (2004). Price discrepancy, option volume, and stock volume all have positive autocorrelations and the autocorrelations decay more slowly in lags than the two order imbalance variables.

## **5. Results**

### **5.1. Sources of Price Impact from Option Trading**

We report the slope coefficients from Fama and MacBeth (1973) regression of equation 5 with Newey and West (1987) adjusted t-statistics to three lags in table 3. The dependent variable is one-day ahead risk adjusted return in columns 1 to 8. The first five columns report univariate regression results. In column 1, we find stock order imbalance surprisingly has little predictive ability as the estimated coefficient is 0.011 with t-statistics of 0.42. We do not find stock order flows contain price information for stocks with options. In column 2, option order imbalance has a slope coefficient of 0.539 with t-statistics of 5.94, suggesting that large buy pressure of delta in the options market is associated with positive future returns. It can be seen that one basis point increase in delta order imbalance will increase the expected stock return by about 0.539 basis point. The difference in informativeness of stock order imbalance and option order imbalance

leads us to think whether the overall stock market order imbalance has predictive ability of the next day's stock return. As shown in column 3, the estimated coefficient of total order imbalance is 0.004 with t-statistics of 1.57. Since option order imbalance is generally much smaller than stock order imbalance, the result is as expected. It is clear that noise of orders submitted by pure stock market investors impedes the information in option order imbalance and makes the overall stock order imbalance uninformative for optionable stocks. We are interested to see whether stock order imbalance loses predictive ability because of option trading. In a separate test, we regress the next day's risk adjusted stock returns on total order imbalance in a sample of all stocks without options during the same period. For these stocks, stock order imbalance is the same as total order imbalance. The result is reported in column 4. For non-optionable stocks, stock order imbalance can significantly predict future returns as the t-statistics reaches -5.09. The results suggest that option trading breaks the information linkage between stock order flows and future stock returns. We suspect informed traders migrate to the options market when options are available. As a result, the information content in stock order flows becomes weaker. Finally column 5 shows price discrepancy in the two markets has a strongly positive coefficient of 1.902 with t-statistics of 7.85. The fact that price discrepancy has a significantly positive coefficient indicates that stock prices tend to move up when the option implied price is higher than the spot price. We plot the time series of t-statistics from the univariate regressions in figure 2.

[Figure 2 about here.]

Next we turn to multivariate results. First we regress the next day's risk adjusted return on both stock and option order imbalance and report the results in column 6 of table 3. It can be seen that consistent with univariate results, option order imbalance retains the positive and significant coefficient and stock order imbalance fails to predict future returns. The result suggests that order imbalances in the two markets have different pricing effects on stock returns. Therefore, it is unlikely that the price impact of option trading is realized through the mechanical price pressure from option market makers' hedging activity. We add price discrepancy as another independent variable in column 7. As can be seen, price discrepancy has a positive and significant coefficient of 1.885 with t-statistics of 7.75 and has little impact on the other two variables.

The full specification with microstructure control variables is tested in column 8. The slope coefficients of stock order imbalance, option order imbalance, and price discrepancy do not change much from previous results. Stock order imbalance has an insignificant coefficient with t-statistics of 1.49. Option order im-

balance has a coefficient of 0.537 with t-statistics of 5.71. The significant predictive ability of option order imbalance supports hypothesis 3 that option trading has *additional* information about stock returns. The coefficient of price discrepancy is 1.861 with t-statistics of 7.64, consistent with hypothesis 4. Therefore, we identify two separate channels of information transmission from the options market to the stock market through both option trades and quotes. Bid-ask spread on the stock market is negatively correlated with future returns and the coefficient is significant with t-statistics of -2.3. Stock turnover does not have predictive ability. Past five-day return is also a significant predictor of future returns with t-statistics of -4.07.

To investigate whether the price impact of option trading is a transitory effect, we use two-day ( $k = 2$ ) and three-day ( $k = 3$ ) ahead risk adjusted returns as dependent variables in estimating equation 5 and report results in columns 9 and 10. Option order imbalance has insignificant coefficients in both columns, suggesting the information in option transactions disseminates quickly. However, price discrepancy has positive and significant coefficients in both columns, suggesting the predictive ability of option quotes carry into longer future than option trades. We repeat the same analysis with  $k$  up to 20. Figure 3 plots the estimated coefficients and 95% confidence intervals for option order imbalance and price discrepancy. Stock order imbalance always has insignificant coefficients and is thus neglected. It is clear that the predictability of option order imbalance is confined to one day ahead. However, price discrepancy is able to significantly predict future returns up to 18 days ahead. The important implication is that information from the options market is permanent rather than transitory as there is no reversal effect detected for option order imbalance and price discrepancy. Therefore, we reject hypothesis 1 but fail to reject hypotheses 2 and 3.

[Figure 3 about here.]

## 5.2. Robustness

There may be concern that during our sample period the market has experienced a major financial crisis in 2008 that can distort the generality of our results. For robustness check, we reproduce previous results in the subsample from 2009 to 2010 only. The results are reported in table 4. It is clear that the subsample results are consistent with table 3 although the magnitude and significance of coefficients decline slightly. Option order imbalance and price discrepancy remain significant predictors of future returns while stock order imbalance is not.

### 5.3. Level of Information Asymmetry

We have shown that both option trades and quotes contain permanent information about future stock prices. If the predictability is due to information asymmetry in the marketplace, the information content should increase in concentration of informed traders. To test hypothesis 4, we use firm size as a proxy for information asymmetry as small firms are usually less transparent than large firms and can attract more informed traders. We re-estimate equation 5 in three size groups based on NYSE breakpoints and report results in table 5. Since the price impact of option trading is the strongest on the next trading day and decays afterward, we only use  $t + 1$  returns as dependent variable when investigating conditional price impact hereafter unless otherwise specified.

Panel A presents the mean and standard deviation of stock order imbalance, option order imbalance, and price discrepancy in the three groups. Option order imbalance has the largest mean and standard deviation in the small firm group. Medium size firms have smaller mean but larger standard deviation of option order imbalance than large firms. For both stock order imbalance and price discrepancy, mean and standard deviation decrease in firm size. It can be seen that both stock and option trading processes are more imbalanced and more volatile for small firms.

Fama-MacBeth regression results are reported in Panel B of table 5. The number of observations in each group is consistent with the distribution of firm size in NYSE population, indicating that there does not exist firm size bias in our sample. In all three size groups, option order imbalance and price discrepancy are significantly and positively related to future returns and stock order imbalance is insignificant. On time series average, one standard deviation increase in option order imbalance is expected to increase future stock return by 3.03 bp for small firms, 2.09 bp for medium size firms, and 2.17 bp for large firms. One standard deviation increase in price discrepancy is expected to increase future stock return by 15.05 bp for small firms, 3.65 bp for medium size firms, and 3.31 bp for large firms. It is clear that small firms have the largest price sensitivity to information in the options market, consistent with our hypothesis 5 that the concentration of informed traders increases the price impact of option trading. Large and medium firms, however, do not show much difference in this price sensitivity.

## 5.4. Short Sale Constraint

We investigate the relation between informational benefit of option trading and short sale constraint in this subsection. Institutional ownership, as a proxy for market supply of short interest, is expected to be negatively related to the difficulty of short selling. Hypothesis 5 therefore conjectures that the price impact of option trading will be larger for firms that are widely held because informed traders with negative information are more likely to be forced to trade options to build a short position. We divide the full sample into three ownership groups based on the percentage of institutional ownership complied from 13-F filings. Panel A of table 6 presents the mean and standard deviation of stock order imbalance, option order imbalance, and price discrepancy. Across ownership groups, all three variables have the largest mean and standard deviation in the low ownership group. Differences in mean and variance between medium and high ownership groups are small.

We report the Fama-MacBeth regression results of equation 5 for the three ownership groups in panel B. It is noticed that option order imbalance has decreasing coefficients and t-statistics as institutional ownership increases but remains a significant predictor in all three groups. One standard deviation increase in option order imbalance is associated with an increase of stock return by 9.31 bp for firms with low institutional ownership, 0.92 bp for firms with medium institutional ownership, and 1.56 bp for firms with high institutional ownership. Stock order imbalance has no predicting power in small and medium ownership groups. However, it has a surprisingly negative and significant coefficient (-0.085) in large ownership group with t-statistics of -2.31. Price discrepancy has the smallest t-statistics (5.49) in high ownership group. One standard deviation increase in price discrepancy is associated with an increase of stock return by 23.33 bp in low ownership group, 17.19 bp in medium ownership group, and 8.31 bp in high ownership group. For both option order imbalance and price discrepancy, the economic significance decreases in institutional ownership, supporting the hypothesis that the informational benefit of option trading is larger for firms with more short sale constraint.

## 5.5. Information v.s. Illiquidity

In this subsection, we investigate the impact of market activeness on information content conveyed through trading process. If the pricing effect of option trading is a spurious result driven by option market illiquidity, we would expect the pricing effect to disappear when trading becomes more active. On the other hand, if

the pricing effect represents arrival of private information, we would expect the pricing effect to increase when trading volume increases because active markets offer low transaction costs to informed traders. To test hypothesis 6, on each trading day in the sample, we sort sample firms into three option volume groups and three stock volume groups separately. Panel A of table 7 presents the mean and standard deviation for each variable in the subsamples. Across option volume groups, it can be seen that when the options market becomes more active: (1) stock order flow becomes more balanced but the standard deviation of stock order imbalance increases; (2) both mean and standard deviation of option order imbalance increase; (3) price discrepancy has larger mean but smaller standard deviation. Across stock volume groups, we find when the stock market becomes more active: (1) the mean of stock order imbalance reduces but the standard deviation increases; (2) the standard deviation of option order imbalance increases but the mean distribution presents a hump shape and peaks at the middle; (3) the mean of price discrepancy reduces and the standard deviation presents a V shape pattern. In summary, option order flow becomes more imbalanced when option trading is active but less so when stock trading is active. Stock trading becomes more balanced when either option or stock trading is active.

We report Fama-MacBeth regression results of equation 5 in the six volume groups in panel B. Across option volume groups, option order imbalance loses predictive ability in the low volume group as the coefficient becomes insignificant. In medium and high option volume groups, option order imbalance has positive and significant coefficients. One standard deviation increase in option order imbalance is associated with an increase of stock return by 2.1 bp and 5.61 bp, respectively. It is clear the activeness of options market has a positive and significant impact on the information content conveyed through option order imbalance. Stock order imbalance has a significant coefficient in medium option volume group only. Price discrepancy has positive and significant coefficients in all three option volume groups. After comparing the economic significance, we find the predictive ability of price discrepancy is not affected much by option trading volume. Across stock volume groups, stock order imbalance has a positive and significant coefficient (0.175) with t-statistics of 2.75 in low stock volume group. However, the predictive ability does not increase when the stock market becomes more active. In both medium and high stock volume groups, stock order imbalance fails to predict future returns. Option order imbalance is always a significant predictor of future returns and the marginal contribution of option order imbalance is large and persistent around 2.8 bp per one standard deviation of increase. Price discrepancy also has large and significant coefficients regardless of stock trading volume. The results uncover both illiquidity and information effects. We find that stock order imbalance has positive correlation with future returns only when option and stock volumes are low. On the other hand,

the predictive ability of option order imbalance increases in options market liquidity, suggesting market activeness reinforces the informational role of options.

## 6. Further Analysis

The previous section has documented that both option trades and quotes contain price information about the underlying stocks. In this section, we perform additional analysis to enhance the understanding on this informational linkage.

### 6.1. Nonlinear Price Impact

We investigate nonlinearity in the relation between order imbalance, price discrepancy, and stock returns in this subsection. Our benchmark specification is the full specification in equation 5. Stock order imbalance, option order imbalance, and price discrepancy all have long and fat tails in distributions. It is possible that stock price sensitivity to these variables will be different in the tails. Our first analysis of nonlinear price impact examines the following specification:

$$Ret_{i,t+1} = \alpha + \beta_1 SOI_{i,t} + \beta_2 OOI_{i,t} + \beta_3 PD_{i,t} + \beta_4 SOI_{i,t}^2 \cdot S1_{i,t} + \beta_5 OOI_{i,t}^2 \cdot S2_{i,t} + \beta_6 PD_{i,t}^2 \cdot S3_{i,t} + \theta X_{i,t} + \epsilon_{i,t+1}, \quad (6)$$

where  $S_1$ ,  $S_2$ , and  $S_3$  are dummy variables that equal to 1 when  $SOI$ ,  $OOI$ , and  $PD$  are positive, respectively, and -1 otherwise. The interaction terms quadratically amplify the contribution of these three independent variables and will parse out the tail effect. We run Fama-MacBeth regression to estimate equation 6 and report results in the first column of table 8. As can be seen, option order imbalance and price discrepancy still have positive and significant coefficients. The surprising result is now stock order imbalance also has a positive and significant coefficient 0.118 with t-statistics of 2.89. The control variables do not seem to be affected. The quadratic terms all have negative coefficients, indicating the price impact is weaker in the tails of order imbalance and price discrepancy.

Next we investigate if there exists asymmetric impact for the three main independent variables by esti-

mating the following equations:

$$\begin{aligned} Ret_{i,t+1} = & \alpha + \beta_1 SOI_{i,t} + \beta_2 OOI_{i,t} + \beta_3 PD_{i,t} + \beta_4 D1_{i,t} + \beta_5 SOI_{i,t} \cdot D1_{i,t} + \beta_6 D2_{i,t} \\ & + \beta_7 OOI_{i,t} \cdot D2_{i,t} + \beta_8 D3_{i,t} + \beta_9 PD_{i,t} \cdot D3_{i,t} + \theta X_{i,t} + \epsilon_{i,t+1}, \end{aligned} \quad (7)$$

where  $D_1$ ,  $D_2$ , and  $D_3$  are binary sign dummy variables for stock order imbalance, option order imbalance, and price discrepancy that is equal to 1 when the associated independent variable is positive and 0 otherwise. The results are reported in the second column of table 8. Stock order imbalance has a negative and significant coefficient -0.12 with t-statistics of -2.35. There is no significant effect on the intercept but the predictive ability is offset when stock order imbalance turns positive. Option order imbalance has a coefficient of 0.614 with t-statistics of 4.17. There is a jump in expected returns around zero option order imbalance as  $\beta_8$  is estimated to be 3.526 with t-statistics of 4.8. However, the interaction term of  $D_2$  and option order imbalance is negative and significant, indicating the return sensitivity to option order imbalance is stronger when there is net sell pressure on the options market. Price discrepancy has a coefficient of 1.618 with t-statistics of 3.95. A jump in intercept is also observed when price discrepancy becomes positive. However, there is no asymmetric return sensitivity as  $\beta_9$  is insignificantly different from zero.

## 6.2. Moving Average and Shocks

It has been shown that stock order imbalance, option order imbalance, and price discrepancy all have positive autocorrelations. In this subsection, we decompose each of these independent variables into a moving average component and a shock component to further investigate the sources of price impact. For example,

$$OOI = OOI_{MA_k} + OOI_{shock_k}, \quad (8)$$

where  $OOI_{MA_k}$  is the average option order imbalance from previous  $k$  trading days. For simplicity, we omit firm and time subscription in equation 8. The same decomposition is also performed for stock order imbalance and price discrepancy. Then we estimate the following equation using Fama-MacBeth regression:

$$Ret_{i,t+1} = \alpha + \beta_1 SOI_{MA_{i,t}} + \beta_2 SOI_{shock_{i,t}} + \beta_3 OOI_{MA_{i,t}} + \beta_4 OOI_{shock_{i,t}} + \beta_5 PD_{MA_{i,t}} + \beta_6 PD_{shock_{i,t}} + \theta X_{i,t} + \epsilon_{i,t+1}, \quad (9)$$

We report results from 3-day, 5-day, and 10-day moving averages decomposition in table 9. Empirical results of longer moving averages are not different thus unreported. Panel A presents the time series average of daily cross sectional correlations between moving average variables and shock variables. The correlations are large and negative for the two order imbalance variables, suggesting a mean-reversion pattern in order flow processes. The 3-day correlations are -0.442 and -0.5 for stock and option order imbalances, respectively. As the length of moving average increases, the correlations become weaker. The correlations are still negative but much smaller for price discrepancy, suggesting price discrepancy is more persistent than order imbalances. The 3-day correlation is only -0.06.

Panel B reports the estimation results of equation 9. The first column reports the results using 3-day moving average decomposition. Although moving average components and shock components are negatively correlated, they are all positively correlated with future stock returns with strong statistical significance. Moving average of stock order imbalance is also marginally significant ( $T=1.87$ ) and the shock component has no predictive ability. The 3-day moving average of option order imbalance has a coefficient of 0.589 with t-statistics of 3.33 and the 3-day shock of option order imbalance has a coefficient of 0.504 with t-statistics of 4.9. In terms of economic significance, one standard deviation increase in 3-day moving average of option order imbalance is associated with an increase of stock return by 2.7 bp and one standard deviation increase in 3-day shock is associated with an increase of stock return by 4.68 bp. For price discrepancy, the marginal contribution is 3.63 bp from moving average and 15.71 bp from shock. Both moving average and shock contains price information in option order imbalance and price discrepancy. But the shocks have larger statistical and economic significance than moving averages. The results are consistent when 5-day or 10-day moving average decomposition is used as shown in the other two columns.

### 6.3. Portfolio Analysis

Since a significant amount of information is conveyed in the options market about future stock returns, it is interesting to see if the informational linkage implies trading profit. In this subsection, we exploit the profitability of order imbalance and price discrepancy by forming long-short portfolios. On every trading day in the sample, we sort firms into quintile portfolios based on stock order imbalance, option order imbalance, and price discrepancy separately. All portfolios are rebalanced the next day at market close. We report close-to-close returns in table 10. Because we measure stock and option order imbalance at 3:55 pm every day, trading strategies on order imbalance variables will have five minutes to be executed. Execution

of the trading strategy on price discrepancy is more difficult as the measure is taken at market close. The first column in table 10 presents the average raw returns of quintile portfolios on stock order imbalance. As can be seen, the quintile portfolio returns present a V shape pattern. The raw return is the lowest on the third quintile portfolio. The long-short strategy does not generate significant return. The second column presents the average returns on quintile portfolios based on option order imbalance. Portfolio returns increase almost monotonically across quintile portfolios. The lowest option order imbalance portfolio has an average return of 0.068 bp and the highest option order imbalance portfolio has an average return of 9.286 bp per day. The trading strategy longing the highest and shorting the lowest option order imbalance portfolios generates an average daily return of 9.217 bp with t-statistics of 5.66. The return difference is significant even after controlling for the three risk factors in Fama-French (1993) and momentum risk. The annual Sharpe ratio of the long-short strategy reaches 3.54. The third column shows that the trading strategy based on price discrepancy is even more profitable. The lowest quintile portfolio has a negative average daily return of -8.578 bp and the highest quintile portfolio has a daily average return of 25.492 bp. The long-short strategy generates an average daily return of 34.07 bp with t-statistics of 12.9. The FF3- and FF4-factor adjusted returns are still over 33 bp per day with t-statistics greater than 12. The annual Sharpe ratio of this strategy reaches 8.23. The cumulative abnormal returns of these three trading strategies are plotted in figure4. During the sample period, trading strategies based on option order imbalance and price discrepancy persistently generate abnormal returns, regardless of the market conditions. The 29-month rate of returns are 75% and 689% on option order imbalance and price discrepancy strategies, respectively.

[Figure 4 about here.]

## 7. Conclusion

Despite the rich empirical evidence of price predictability from option trades and quotes, little is known about how the price impact is realized. In this article, we investigate the sources of feedback effect from the options market conditioning on trading activities. By decomposing stock order flows into an option market makers' delta hedging component and a pure stock market initiated component, we are able to test whether the options market convey price information through option trades and quotes in addition to price information revealed in the stock market. We find stock order imbalance significantly predicts future returns only for stocks without options. In the sample of optionable stocks, however, stock order imbalance reveals no

price information. On the other hand, option order imbalance has strong predictive ability of future stock returns in the cross section. It is likely that informed traders migrate to the options market once options become available. Other than option transactions, we find option quotes are even more informative about future price movement. Both option trades and quotes have permanent price impact and no reversal effect is observed in longer horizons. Price information in option order imbalance dies out after two days while price discrepancy can carry information into three week later. The results are robust in post-crisis periods. We then investigate the link between the information content in the options market and firm size, institutional ownership, and market liquidity. Consistent with theoretical predictions, we find the price impact of both option trades and quotes is stronger for firms that are small and widely held. The concentration of informed traders and the constraint of short selling enhances the informational benefit of option trading in price discovery. The options market also becomes more informative when option trading is active, consistent with the prediction by Easley, O'Hara, and Srinivas (1998) that informed trading is more likely to happen in deep and liquid markets. Further analysis shows that option order imbalance and price discrepancy become noisy in tails. Price sensitivity to option order imbalance is asymmetric as the impact of negative imbalance has a larger impact on the underlying price. We also document significant abnormal returns from trading strategies based on option market signals.

Our findings indicate that the options market plays a key role in embedding private information into stock prices. The imbalance in option order flows and stock order flows can contain different messages for the market. In fact, these two order imbalances are negatively correlated on average. We have also identified both option trades and quotes as separate information transmission channels from the options market to the stock market. Option order imbalance does not necessarily lead to a large discrepancy between option implied price and spot price. The results suggest that there exist market frictions preventing all investors utilizing information in option order flows. For example, the cost of acquiring such information can be too high for most investors but not for option market makers. It is also possible that option order flows are too complex for some investors to understand. As a result, there is an information segmentation in the option implied and actual prices. We do not claim that the option implied price is more efficient than the actual price because option market investors can also miss information in the stock market if cost of stock market information is high for them. We are also conservative on whether option trading improves information efficiency compared to a single security market. These questions call for more empirical analysis in future.

The proposed option order imbalance focuses on delta risk and has shown significant predictability

about the underlying market. Actually option contracts have other risk exposures that can be analyzed in similar ways. It would be interesting to investigate the informational link between derivative markets and the underlying market in a multidimensional framework because some risk can only be traded in the options market. Understanding this question is important to both academicians and practitioners. Future research on how to incorporate the microstructure impact into option pricing is needed.

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**Table 1****Summary statistics of the trade sample**

The table reports statistics of options market activity between April 2008 and August 2010. Option transaction data with trade direction (buyer or seller initiated), implied volatility, and corresponding spot price is provided by Trader Alert LLC. All trade records except canceled trades are included in the sample.  $\delta$  is calculated using Black-Scholes (1973) formula assuming 0 interest rate and dividend rate.  $T$  is days to expiry. Options exchange code: AMEX - American Stock Exchange, PCX - New York Stock Exchange ARCA, BATS (launched on February 26, 2010), BOX - Boston Options Exchange, CBOE - Chicago Board Options Exchange, ISE - International Securities Exchange, NASDAQ, and PHLX - Philadelphia Stock Exchange.

| Statistics                                                                        | All        | Buy Call  | Sell Call | Buy Put   | Sell Put  |
|-----------------------------------------------------------------------------------|------------|-----------|-----------|-----------|-----------|
| <i>Panel A Number of firms:</i>                                                   |            |           |           |           |           |
| Average number of firms per day                                                   | 2605       | 2110      | 2127      | 1796      | 1775      |
| Max number of firms per day                                                       | 2933       | 2435      | 2517      | 2421      | 2259      |
| Min number of firms per day                                                       | 2225       | 1630      | 1653      | 1286      | 1314      |
| Std number of firms per day                                                       | 112        | 145       | 141       | 143       | 120       |
| <i>Panel B Trade and trade size statistics:</i>                                   |            |           |           |           |           |
| Mean trade size                                                                   | 23.31      | 21.99     | 20.55     | 25.54     | 24.58     |
| Mean daily number of trades                                                       | 539,912    | 138,540   | 140,345   | 98,436    | 95,118    |
| Mean daily trade volume                                                           | 12,584,280 | 3,046,000 | 2,884,506 | 2,514,197 | 2,338,109 |
| Mean daily premium (in millions)                                                  | 3,023.81   | 709.26    | 664.38    | 628.06    | 584.09    |
| <i>Panel C Percentages of trade records classified into different categories:</i> |            |           |           |           |           |
| $OTM :  \delta  < 37.5\%$                                                         | 50.37      | 10.85     | 10.33     | 11.46     | 10.55     |
| $ATM : 37.5\% \leq  \delta  \leq 62.5\%$                                          | 32.81      | 8.54      | 8.13      | 6.24      | 5.59      |
| $ITM :  \delta  > 62.5\%$                                                         | 16.82      | 4.82      | 4.46      | 2.28      | 2.44      |
| $T \leq 30$ days                                                                  | 45.69      | 11.28     | 10.43     | 9.07      | 8.57      |
| $31 < T \leq 60$ days                                                             | 25.10      | 5.92      | 5.54      | 5.25      | 4.73      |
| $T > 60$ days                                                                     | 29.21      | 7.00      | 6.96      | 5.66      | 5.27      |
| <i>Panel D Percentages of trade volume by exchange:</i>                           |            |           |           |           |           |
| AMEX                                                                              | 8.37       | 1.97      | 1.76      | 1.76      | 1.66      |
| BATS                                                                              | 0.09       | 0.02      | 0.02      | 0.02      | 0.02      |
| BOX                                                                               | 3.95       | 0.96      | 0.92      | 0.9       | 0.87      |
| CBOE                                                                              | 26.45      | 6.68      | 6.26      | 5.55      | 5.08      |
| ISE                                                                               | 26.79      | 6.24      | 5.99      | 5.46      | 5.08      |
| NASDAQ                                                                            | 2.55       | 0.59      | 0.59      | 0.54      | 0.53      |
| PCX                                                                               | 12.76      | 2.73      | 2.66      | 2.4       | 2.26      |
| PHLX                                                                              | 19.04      | 5.02      | 4.73      | 3.37      | 3.11      |

**Table 2****Descriptive statistics of main variables**

Panel A reports the time series average of cross sectional statistics of individual stock's daily four-factor adjusted returns (Ret), stock order imbalance (SOI), option order imbalance (OOI), absolute ratio of option order imbalance and stock order imbalance ( $|OOI/SOI|$ ), price discrepancy (PD), option volume, and stock volume. The filtering excludes (i) off-hour trades, (ii) trades within first 15 minutes after market open and last 5 minutes before market close, (iii) late and canceled trades, (iv) data error, and (v) unclassified trades. Ret, SOI, and OOI are reported in basis points and  $|OOI/SOI|$  and PD in percentage.  $|OOI/SOI|$  is winsorized at top and bottom 0.5%. Option volume is the natural logarithm of the number of option lots traded. Stock volume is the natural logarithm of hundreds of stock shares traded. Panel B reports the time series average of daily cross sectional correlations. Panel C reports the cross sectional average of autocorrelations of each variable up to five lags.

Panel A Descriptive Statistics:

| Variables     | N       | Mean   | Std     | Median | Min       | Max      | Skew   | Kurt    |
|---------------|---------|--------|---------|--------|-----------|----------|--------|---------|
| Ret           | 1390056 | -0.786 | 334.68  | -7.945 | -2957.288 | 4282.336 | 1.510  | 56.233  |
| SOI           | 1390305 | -0.799 | 26.399  | -0.343 | -516.63   | 429.67   | -2.030 | 258.579 |
| OOI           | 1390305 | -0.026 | 6.842   | -0.006 | -140.624  | 132.127  | -0.255 | 333.446 |
| OOI/SOI       | 1387967 | 44.685 | 162.646 | 6.742  | 0.001     | 1702.875 | 7.657  | 67.575  |
| PD            | 1390305 | -0.156 | 4.414   | -0.127 | -67.483   | 126.173  | 6.418  | 570.826 |
| Option volume | 1390305 | 5.336  | 2.43    | 5.279  | 0.005     | 13.043   | 0.086  | -0.456  |
| Stock volume  | 1387972 | 8.899  | 1.455   | 8.823  | 3.76      | 14.856   | 0.223  | 0.111   |

Panel B Correlation:

|               | Ret   | SOI    | OOI    | TOI    | PD     | Vol <sub>op</sub> | Vol <sub>stk</sub> |
|---------------|-------|--------|--------|--------|--------|-------------------|--------------------|
| Ret           | 1     |        |        |        |        |                   |                    |
| SOI           | 0.000 | 1      |        |        |        |                   |                    |
| OOI           | 0.008 | -0.173 | 1      |        |        |                   |                    |
| TOI           | 0.002 | 0.958  | 0.084  | 1      |        |                   |                    |
| PD            | 0.018 | -0.006 | -0.001 | -0.007 | 1      |                   |                    |
| Option volume | 0.000 | 0.013  | 0.002  | 0.014  | -0.005 | 1                 |                    |
| Stock volume  | 0.005 | 0.011  | 0.007  | 0.013  | 0.018  | 0.704             | 1                  |

Panel C Autocorrelation:

| lag | Ret    | SOI   | OOI   | PD    | Vol <sub>op</sub> | Vol <sub>stk</sub> |
|-----|--------|-------|-------|-------|-------------------|--------------------|
| 1   | -0.021 | 0.094 | 0.033 | 0.210 | 0.356             | 0.289              |
| 2   | -0.012 | 0.047 | 0.012 | 0.185 | 0.305             | 0.253              |
| 3   | -0.015 | 0.034 | 0.007 | 0.169 | 0.275             | 0.233              |
| 4   | -0.004 | 0.027 | 0.004 | 0.146 | 0.256             | 0.220              |
| 5   | -0.011 | 0.024 | 0.005 | 0.122 | 0.244             | 0.212              |

Table 3

**Predicting stock returns with order imbalance and price discrepancy**

The table reports Fama-MacBeth (1973) regression results of equation 5. The dependent variable is the risk adjusted stock return on day  $t + 1$  in columns 1-8, on day  $t + 2$  in column 9 and day  $t + 3$  in column 10. All independent variables are measured at day  $t$ . At 15:55 EST every day, we measure option order imbalance (OOI), stock order imbalance (SOI), and total order imbalance (TOI) using trade information from 9:45 EST on the same day. All other independent variables are measured at market close every day. PD is the difference of implied stock price from ATM one-month call and put options minus the spot price scaled by the spot price. Spread is the difference between closing bid and ask prices on the stock market. Turnover is the total number of shares traded scaled by number of shares outstanding.  $Ret_{5,-1}$  is the raw return from day  $t - 5$  to day  $t - 1$ . T-statistics are reported in parentheses.

|              | (1)               | (2)               | (3)               | (4)               | (5)               | (6)               | (7)               | (8)                | (9)                | (10)               |
|--------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|--------------------|--------------------|--------------------|
| intercept    | -0.901<br>(-0.82) | -0.809<br>(-0.73) | -0.896<br>(-0.81) | 3.709<br>(1.32)   | -0.374<br>(-0.33) | -0.905<br>(-0.83) | -0.492<br>(-0.44) | -2.057<br>(-1.19)  | -1.887<br>(-1.42)  | -2.337<br>(-2.10)  |
| SOI          | 0.011<br>(0.42)   |                   |                   |                   |                   | 0.034<br>(1.17)   | 0.042<br>(1.42)   | 0.043<br>(1.49)    | -0.037<br>(-1.44)  | 0.012<br>(0.59)    |
| OOI          |                   | 0.539<br>(5.94)   |                   |                   |                   | 0.588<br>(6.07)   | 0.596<br>(6.15)   | 0.537<br>(5.71)    | -0.052<br>(-0.54)  | -0.103<br>(-1.29)  |
| TOI          |                   |                   | 0.004<br>(1.57)   | -0.002<br>(-5.09) |                   |                   |                   |                    |                    |                    |
| PD           |                   |                   |                   |                   | 1.902<br>(7.85)   |                   | 1.885<br>(7.75)   | 1.861<br>(7.64)    | 0.584<br>(3.27)    | 0.544<br>(3.33)    |
| spread       |                   |                   |                   |                   |                   |                   |                   | -18.849<br>(-2.30) | -11.686<br>(-1.65) | -17.516<br>(-2.64) |
| turnover     |                   |                   |                   |                   |                   |                   |                   | -0.024<br>(-0.05)  | -0.517<br>(-1.39)  | -0.093<br>(-0.27)  |
| $Ret_{5,-1}$ |                   |                   |                   |                   |                   |                   |                   | -0.006<br>(-4.07)  | -0.006<br>(-4.20)  | -0.006<br>(-4.13)  |
| N of obs     | 1390056           | 1390056           | 1387728           | 1697326           | 1390056           | 1390056           | 1390056           | 1389920            | 1389685            | 1389450            |
| Adj R-sq     | 0.0022            | 0.0013            | 0.0023            | 0.0002            | 0.0028            | 0.0035            | 0.0063            | 0.0177             | 0.0129             | 0.0118             |

Table 4

**Robustness in subsample period of 2009-2010**

The table reports Fama-MacBeth (1973) regression results of equation 5 using data from 2009 and 2010 only. The dependent variable is the risk adjusted stock return on day  $t + 1$  in columns 1-8, on day  $t + 2$  in column 9 and day  $t + 3$  in column 10. At 15:55 EST every day, we measure option order imbalance (OOI), stock order imbalance (SOI), and total order imbalance (TOI) using trade information from 9:45 EST on the same day. All other independent variables are measured at market close every day. PD is the difference of implied stock price from ATM one-month call and put options minus the spot price scaled by the spot price. Spread is the difference between closing bid and ask prices on the stock market. Turnover is the total number of shares traded scaled by number of shares outstanding.  $Ret_{5,-1}$  is the raw return from day  $t - 5$  to day  $t - 1$ . T-statistics are reported in parentheses.

|              | (1)               | (2)               | (3)               | (4)               | (5)               | (6)               | (7)               | (8)                | (9)                | (10)               |
|--------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|--------------------|--------------------|--------------------|
| intercept    | -1.274<br>(-1.31) | -1.221<br>(-1.25) | -1.322<br>(-1.37) | 8.226<br>(3.20)   | -0.756<br>(-0.76) | -1.388<br>(-1.45) | -1.004<br>(-1.03) | 1.240<br>(1.00)    | 0.339<br>(0.31)    | -0.555<br>(-0.5)   |
| SOI          | 0.032<br>(0.95)   |                   |                   |                   |                   | 0.047<br>(1.34)   | 0.051<br>(1.46)   | 0.056<br>(1.68)    | -0.045<br>(-1.55)  | 0.008<br>(0.37)    |
| OOI          |                   | 0.480<br>(4.72)   |                   |                   |                   | 0.511<br>(4.59)   | 0.518<br>(4.64)   | 0.487<br>(4.40)    | -0.091<br>(-0.90)  | -0.061<br>(-0.66)  |
| TOI          |                   |                   | 0.006<br>(1.69)   | -0.002<br>(-3.87) |                   |                   |                   |                    |                    |                    |
| PD           |                   |                   |                   |                   | 1.646<br>(5.45)   |                   | 1.619<br>(5.34)   | 1.570<br>(5.20)    | 0.681<br>(2.96)    | 0.657<br>(3.32)    |
| spread       |                   |                   |                   |                   |                   |                   |                   | -21.466<br>(-2.04) | -15.984<br>(-1.84) | -22.088<br>(-2.69) |
| turnover     |                   |                   |                   |                   |                   |                   |                   | -0.860<br>(-1.89)  | -0.994<br>(-2.74)  | -0.691<br>(-2.01)  |
| $Ret_{5,-1}$ |                   |                   |                   |                   |                   |                   |                   | -0.006<br>(-4.25)  | -0.005<br>(-3.80)  | -0.005<br>(-3.62)  |
| N of obs     | 1390056           | 1390056           | 1387728           | 1697326           | 1390056           | 1390056           | 1390056           | 1389920            | 1389685            | 1389450            |
| Adj R-sq     | 0.0025            | 0.0015            | 0.0026            | 0.0002            | 0.0027            | 0.0040            | 0.0067            | 0.0165             | 0.0115             | 0.0101             |

**Table 5****Concentration of informed traders: subsample results in size groups**

Panel A reports the mean and standard deviation of stock order imbalance (SOI), option order imbalance (OOI), and price discrepancy (PD) in three size groups. Each firm is characterized into small size ( $< 30\%$ ), medium size ( $30\% \sim 70\%$ ), large size ( $\geq 70\%$ ) groups using monthly NYSE breakpoints. The dependent variable is the risk adjusted stock return on day  $t + 1$ . All independent variables are measured at day  $t$ . At 15:55 EST every day, we measure OOI and SOI using trade information from 9:45 EST on the same day. All other independent variables are measured at market close every day. PD is the difference of implied stock price from ATM one-month call and put options minus the spot price scaled by the spot price. Panel B reports Fama-MacBeth (1973) regression results of equation 5 for each group with  $k = 1$ . Spread is the difference between closing bid and ask prices on the stock market. Turnover is the total number of shares traded scaled by number of shares outstanding.  $Ret_{-5,-1}$  is the raw return from day  $t - 5$  to day  $t - 1$ . T-statistics are reported in parentheses.

| <u>Panel A Variable Statistics:</u> |              |               |              |
|-------------------------------------|--------------|---------------|--------------|
|                                     | <i>small</i> | <i>medium</i> | <i>large</i> |
| $mean_{SOI}$                        | -2.195       | -0.295        | 0.109        |
| $std_{SOI}$                         | 37.846       | 20.702        | 12.198       |
| $mean_{OOI}$                        | -0.059       | -0.006        | -0.012       |
| $std_{OOI}$                         | 8.408        | 6.040         | 3.939        |
| $mean_{PD}$                         | -0.161       | -0.162        | -0.131       |
| $std_{PD}$                          | 6.055        | 2.962         | 2.021        |

| <u>Panel B Regression Results:</u> |                    |                    |                    |
|------------------------------------|--------------------|--------------------|--------------------|
|                                    | <i>small</i>       | <i>medium</i>      | <i>large</i>       |
| intercept                          | 1.733<br>(0.62)    | -3.243<br>(-1.74)  | -3.252<br>(-1.53)  |
| SOI                                | 0.054<br>(1.25)    | -0.007<br>(-0.20)  | 0.038<br>(0.59)    |
| OOI                                | 0.400<br>(2.45)    | 0.354<br>(2.92)    | 0.843<br>(4.18)    |
| PD                                 | 3.070<br>(7.77)    | 1.750<br>(5.47)    | 2.826<br>(4.72)    |
| spread                             | -50.614<br>(-2.72) | -28.490<br>(-2.38) | -13.156<br>(-1.85) |
| turnover                           | -0.525<br>(-0.93)  | 0.271<br>(0.42)    | -0.205<br>(-0.24)  |
| $Ret_{-5,-1}$                      | -0.006<br>(-4.08)  | -0.005<br>(-2.58)  | -0.010<br>(-4.16)  |
| N of obs                           | 444820             | 563232             | 381868             |

**Table 6****Short sale constraint and the informational benefit from option trading**

Panel A reports the mean and standard deviation of stock order imbalance (SOI), option order imbalance (OOI), and price discrepancy (PD) in three ownership groups. Each firm is characterized into low ( $< 30\%$ ), medium ( $30\% \sim 70\%$ ), and high ( $\geq 70\%$ ) institutional ownership groups based on 13-F filings. The dependent variable is the risk adjusted stock return on day  $t + 1$ . All independent variables are measured at day  $t$ . At 15:55 EST every day, we measure SOI and OOI using trade information from 9:45 EST on the same day. All other independent variables are measured at market close every day. PD is the difference of implied stock price from ATM one-month call and put options minus the spot price scaled by the spot price. Panel B reports Fama-MacBeth (1973) regression results of equation 5 for each group with  $k = 1$ . Spread is the difference between closing bid and ask prices on the stock market. Turnover is the total number of shares traded scaled by number of shares outstanding.  $Ret_{-5,-1}$  is the raw return from day  $t - 5$  to day  $t - 1$ . T-statistics are reported in parentheses.

| <u>Panel A Variable Statistics:</u> |            |               |             |
|-------------------------------------|------------|---------------|-------------|
|                                     | <i>low</i> | <i>medium</i> | <i>high</i> |
| $mean_{SOI}$                        | -1.820     | -0.424        | -0.276      |
| $std_{SOI}$                         | 35.300     | 18.669        | 21.908      |
| $mean_{OOI}$                        | -0.071     | 0.003         | -0.019      |
| $std_{OOI}$                         | 7.907      | 5.205         | 6.154       |
| $mean_{PD}$                         | -0.220     | -0.136        | -0.119      |
| $std_{PD}$                          | 6.511      | 2.785         | 2.065       |

| <u>Panel B Regression Results:</u> |                    |                    |                    |
|------------------------------------|--------------------|--------------------|--------------------|
|                                    | <i>low</i>         | <i>medium</i>      | <i>high</i>        |
| intercept                          | -0.695<br>(-0.26)  | -2.492<br>(-1.25)  | -4.277<br>(-1.99)  |
| SOI                                | 0.056<br>(1.18)    | 0.096<br>(1.85)    | -0.085<br>(-2.31)  |
| OOI                                | 0.727<br>(3.62)    | 0.598<br>(3.38)    | 0.383<br>(3.21)    |
| PD                                 | 2.249<br>(6.04)    | 2.519<br>(5.63)    | 2.859<br>(5.49)    |
| spread                             | -12.053<br>(-1.11) | -43.696<br>(-4.03) | -33.889<br>(-2.16) |
| turnover                           | -0.674<br>(-1.21)  | 1.412<br>(1.45)    | 0.905<br>(1.09)    |
| $Ret_{-5,-1}$                      | -0.010<br>(-5.16)  | -0.004<br>(-2.04)  | -0.004<br>(-2.19)  |
| N of obs                           | 416809             | 556060             | 417051             |

**Table 7****Market activeness and the predictive ability of security trading**

Panel A reports the mean and standard deviation of stock order imbalance (SOI), option order imbalance (OOI), and price discrepancy (PD) in option and stock volume groups. Each firm is characterized into low ( $< 30\%$ ), medium ( $30\% \sim 70\%$ ), and high ( $\geq 70\%$ ) option and stock volume groups separately. The dependent variable is the risk adjusted stock return on day  $t + 1$ . All independent variables are measured at day  $t$ . At 15:55 EST every day, we measure SOI and OOI using trade information from 9:45 EST on the same day. All other independent variables are measured at market close every day. PD is the difference of implied stock price from ATM one-month call and put options minus the spot price scaled by the spot price. Panel B reports Fama-MacBeth (1973) regression results of equation 5 for each group with  $k = 1$ . Spread is the difference between closing bid and ask prices on the stock market. Turnover is the total number of shares traded scaled by number of shares outstanding.  $Ret_{-5,-1}$  is the raw return from day  $t - 5$  to day  $t - 1$ . T-statistics are reported in parentheses.

Panel A Variable Statistics:

|              | <u>Option volume</u> |               |             | <u>Stock volume</u> |               |             |
|--------------|----------------------|---------------|-------------|---------------------|---------------|-------------|
|              | <i>low</i>           | <i>medium</i> | <i>high</i> | <i>low</i>          | <i>medium</i> | <i>high</i> |
| $mean_{SOI}$ | -1.036               | -0.928        | -0.357      | -0.992              | -0.930        | -0.400      |
| $std_{SOI}$  | 18.391               | 23.705        | 32.853      | 12.624              | 23.209        | 36.327      |
| $mean_{OOI}$ | -0.004               | -0.025        | -0.051      | -0.047              | -0.060        | 0.044       |
| $std_{OOI}$  | 0.314                | 2.224         | 12.110      | 4.463               | 6.291         | 7.739       |
| $mean_{PD}$  | -0.103               | -0.178        | -0.177      | -0.250              | -0.151        | -0.062      |
| $std_{PD}$   | 5.349                | 3.373         | 3.042       | 3.849               | 3.517         | 4.583       |

Panel B Regression Results:

|               | <u>Option volume</u> |                   |                    | <u>Stock volume</u> |                    |                    |
|---------------|----------------------|-------------------|--------------------|---------------------|--------------------|--------------------|
|               | <i>low</i>           | <i>medium</i>     | <i>high</i>        | <i>low</i>          | <i>medium</i>      | <i>high</i>        |
| intercept     | -5.917<br>(-2.93)    | -3.397<br>(-1.71) | -0.281<br>(-0.11)  | -6.842<br>(-2.89)   | -0.186<br>(-0.10)  | -0.707<br>(-0.30)  |
| SOI           | -0.024<br>(-0.49)    | 0.107<br>(2.64)   | 0.060<br>(1.27)    | 0.175<br>(2.75)     | 0.019<br>(0.57)    | 0.047<br>(0.92)    |
| OOI           | 1.156<br>(0.55)      | 1.235<br>(4.16)   | 0.515<br>(4.89)    | 0.827<br>(3.61)     | 0.554<br>(4.18)    | 0.543<br>(3.26)    |
| PD            | 2.057<br>(5.56)      | 3.423<br>(6.95)   | 3.244<br>(4.30)    | 3.688<br>(8.14)     | 2.262<br>(5.80)    | 3.496<br>(5.18)    |
| spread        | -8.586<br>(-0.65)    | -19.92<br>(-1.69) | -48.603<br>(-3.54) | -7.898<br>(-0.83)   | -26.085<br>(-2.45) | -59.418<br>(-2.60) |
| turnover      | 4.198<br>(3.37)      | 1.163<br>(1.70)   | -0.648<br>(-1.03)  | 6.034<br>(2.74)     | -0.107<br>(-0.14)  | -0.381<br>(-0.68)  |
| $Ret_{-5,-1}$ | -0.005<br>(-2.46)    | -0.005<br>(-3.00) | -0.007<br>(-3.39)  | -0.003<br>(-1.95)   | -0.005<br>(-2.70)  | -0.009<br>(-4.55)  |
| N of obs      | 416737               | 556271            | 416912             | 418499              | 555284             | 416137             |

**Table 8****Nonlinear price impact from option and stock trading**

The table reports Fama-MacBeth (1973) regression results of equation 6 and equation 7. The dependent variable is the risk adjusted stock return on day  $t + 1$ . All independent variables are measured at day  $t$ . At 15:55 EST every day, we measure option order imbalance (OOI), stock order imbalance (SOI) using trade information from 9:45 EST on the same day. All other independent variables are measured at market close every day. PD is the difference of implied stock price from ATM one-month call and put options minus the spot price scaled by the spot price. Spread is the difference between closing bid and ask prices on the stock market. Turnover is the total number of shares traded scaled by number of shares outstanding.  $Ret_{-5,-1}$  is the raw return from day  $t - 5$  to day  $t - 1$ . T-statistics are reported in parentheses.

|                   |                    |                    |
|-------------------|--------------------|--------------------|
| intercept         | -1.428<br>(-0.83)  | -10.968<br>(-6.00) |
| SOI               | 0.118<br>(2.89)    | -0.120<br>(-2.35)  |
| OOI               | 1.044<br>(6.51)    | 0.614<br>(4.17)    |
| PD                | 5.641<br>(10.72)   | 1.618<br>(3.95)    |
| spread            | -18.060<br>(-2.22) | -14.867<br>(-1.77) |
| turnover          | 0.195<br>(0.40)    | -0.715<br>(-1.44)  |
| $Ret_{-5,-1}$     | -0.006<br>(-3.97)  | -0.006<br>(-3.96)  |
| $S_1 \cdot SOI^2$ | 0.000<br>(-1.42)   |                    |
| $S_2 \cdot OOI^2$ | -0.013<br>(-3.42)  |                    |
| $S_3 \cdot PD^2$  | -0.078<br>(-7.89)  |                    |
| D1                |                    | 0.547<br>(0.54)    |
| D2                |                    | 3.526<br>(4.80)    |
| D3                |                    | 20.198<br>(15.23)  |
| D1*SOI            |                    | 0.298<br>(3.41)    |
| D2*OOI            |                    | -0.445<br>(-2.04)  |
| D3*PD             |                    | -0.011<br>(-0.02)  |

**Table 9****Moving average and shocks in order imbalance and price discrepancy**

Panel A presents the time series average of cross sectional correlations between moving average and shock components for each variable. The table reports Fama-MacBeth (1973) regression results of equation 9 using 3day, 5day, and 10day moving averages. The dependent variable is the risk adjusted stock return on day  $t + 1$ . All independent variables are measured at day  $t$ . At 15:55 EST every day, we measure option order imbalance (OOI), stock order imbalance (SOI) using trade information from 9:45 EST on the same day. All other independent variables are measured at market close every day. PD is the difference of implied stock price from ATM one-month call and put options minus the spot price scaled by the spot price. Spread is the difference between closing bid and ask prices on the stock market. Turnover is the total number of shares traded scaled by number of shares outstanding.  $Ret_{-5,-1}$  is the raw return from day  $t - 5$  to day  $t - 1$ . T-statistics are reported in parentheses.

| <u>Panel A Correlations:</u> |        |        |        |  |
|------------------------------|--------|--------|--------|--|
|                              | 3day   | 5day   | 10day  |  |
| $SOI_{MA}$ and $SOI_{shock}$ | -0.442 | -0.358 | -0.254 |  |
| $OOI_{MA}$ and $OOI_{shock}$ | -0.500 | -0.420 | -0.329 |  |
| $PD_{MA}$ and $PD_{shock}$   | -0.060 | -0.022 | -0.026 |  |

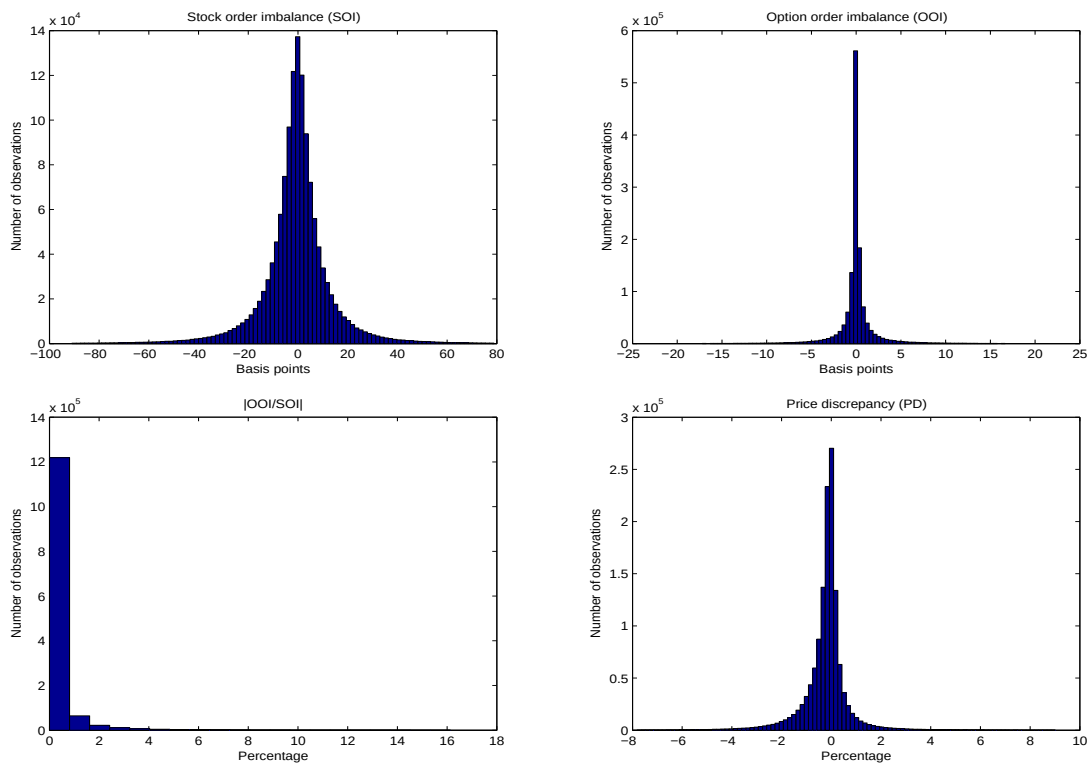
  

| <u>Panel B Regression Results:</u> |                    |                    |                    |  |
|------------------------------------|--------------------|--------------------|--------------------|--|
|                                    | 3day               | 5day               | 10day              |  |
| intercept                          | -2.513<br>(-1.45)  | -2.800<br>(-1.59)  | -3.341<br>(-1.77)  |  |
| $SOI_{MA}$                         | 0.092<br>(1.87)    | 0.057<br>(1.04)    | 0.099<br>(1.41)    |  |
| $SOI_{shock}$                      | 0.043<br>(1.45)    | 0.042<br>(1.37)    | 0.036<br>(1.22)    |  |
| $OOI_{MA}$                         | 0.589<br>(3.33)    | 0.652<br>(3.63)    | 0.736<br>(3.05)    |  |
| $OOI_{shock}$                      | 0.504<br>(4.90)    | 0.568<br>(5.93)    | 0.550<br>(5.56)    |  |
| $PD_{MA}$                          | 1.111<br>(3.81)    | 0.756<br>(1.80)    | 1.077<br>(3.54)    |  |
| $PD_{shock}$                       | 3.406<br>(8.71)    | 3.260<br>(8.15)    | 2.897<br>(8.17)    |  |
| spread                             | -19.799<br>(-2.36) | -15.739<br>(-1.88) | -16.622<br>(-1.94) |  |
| turnover                           | 0.082<br>(0.17)    | 0.148<br>(0.30)    | 0.105<br>(0.22)    |  |
| $Ret_{-5,-1}$                      | -0.007<br>(-4.46)  | -0.007<br>(-4.28)  | -0.007<br>(-4.52)  |  |

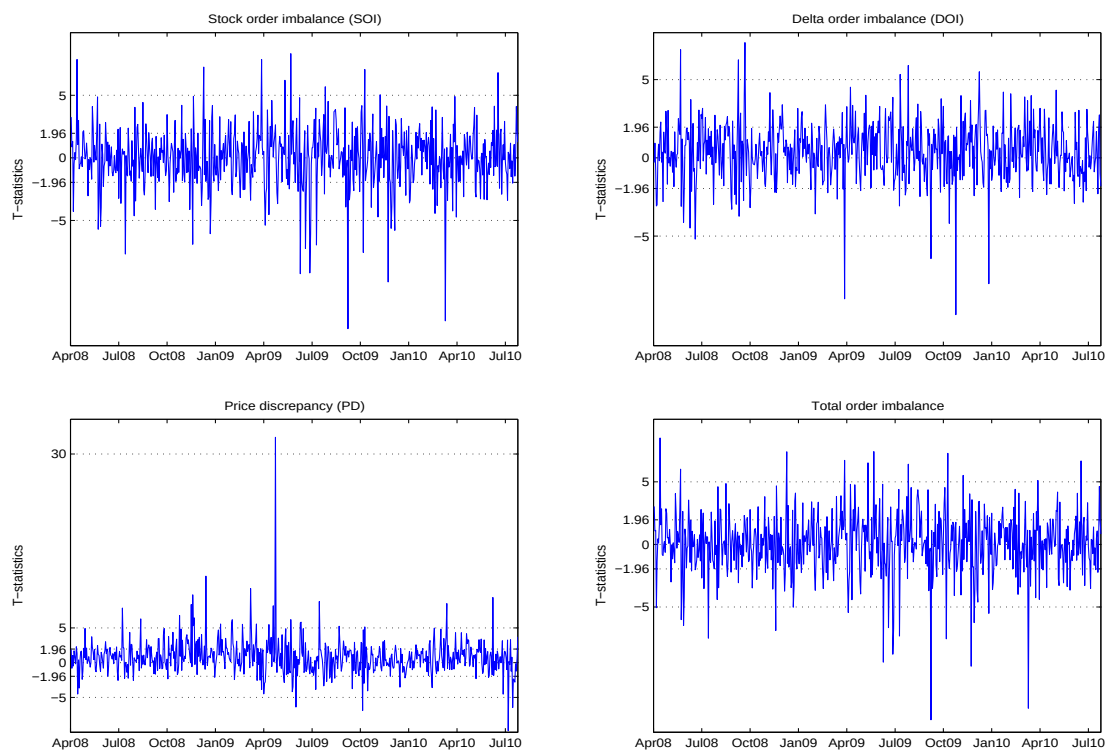
**Table 10****Abnormal returns on order imbalance portfolios**

This table reports the average returns on decile portfolios based on stock order imbalance (SOI), option order imbalance (OOI), and price discrepancy (PD). The portfolios are rebalanced every day at market close based on order imbalance between 9:45 and 15:55 EST. Return differences are also reported with Fama-French (1993) three and four factor adjusted return differences. T-statistics are reported in parentheses.

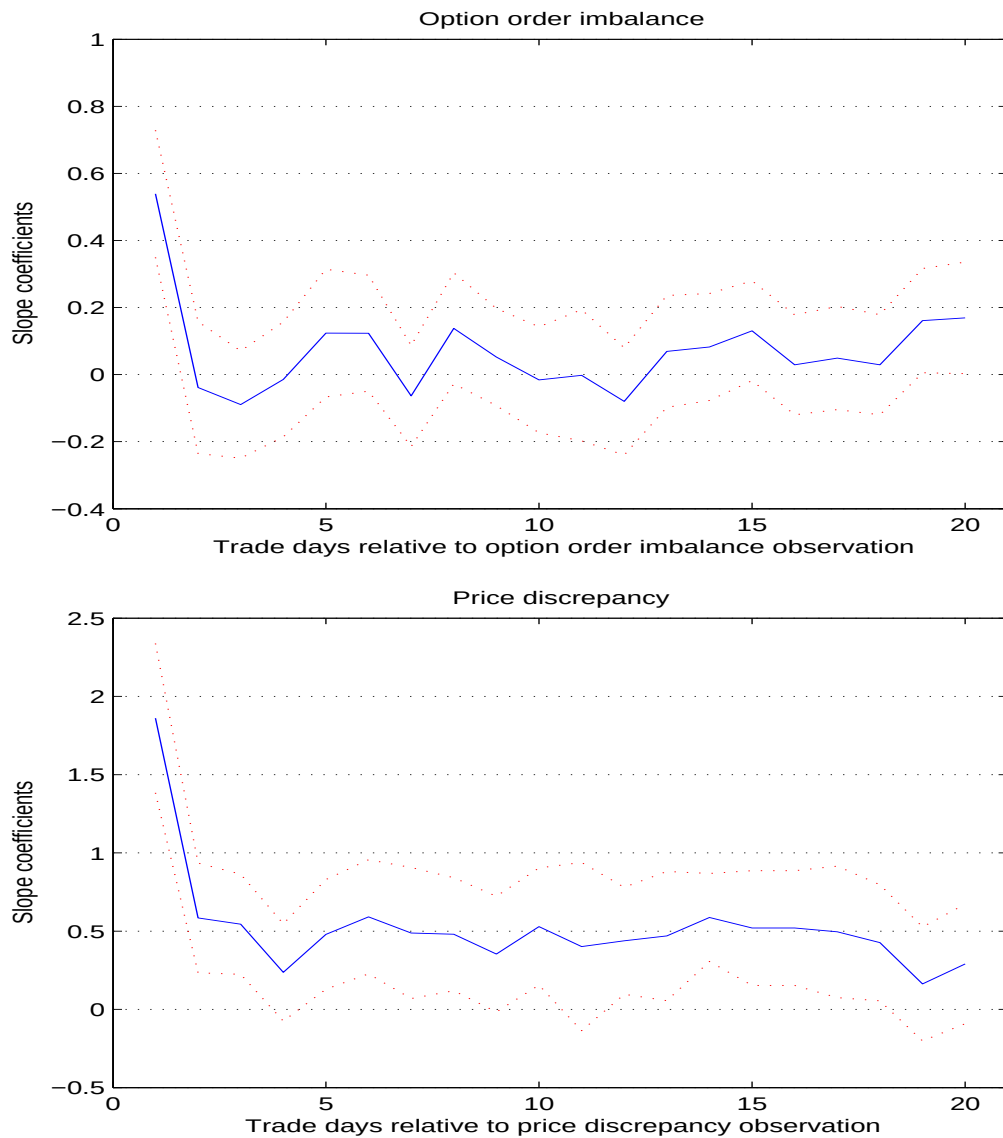
|           | SOI    | OOI    | PD      |
|-----------|--------|--------|---------|
| low - 1   | 6.885  | 0.068  | -8.578  |
| 2         | 1.483  | 3.317  | -3.356  |
| 3         | 1.358  | 2.764  | 0.171   |
| 4         | 2.028  | 5.326  | 7.027   |
| high - 5  | 7.484  | 9.286  | 25.492  |
| 5-1       | 0.599  | 9.217  | 34.070  |
| t-stat    | (0.29) | (5.66) | (12.90) |
| FF3 alpha | 0.573  | 9.245  | 33.992  |
| t-stat    | (0.28) | (5.75) | (12.88) |
| FF4 alpha | 0.233  | 9.065  | 33.767  |
| t-stat    | (0.12) | (5.65) | (12.79) |
| Sharpe    | 0.11   | 3.54   | 8.23    |



**Figure 1**  
Histogram of stock order imbalance, option order imbalance, and price discrepancy.



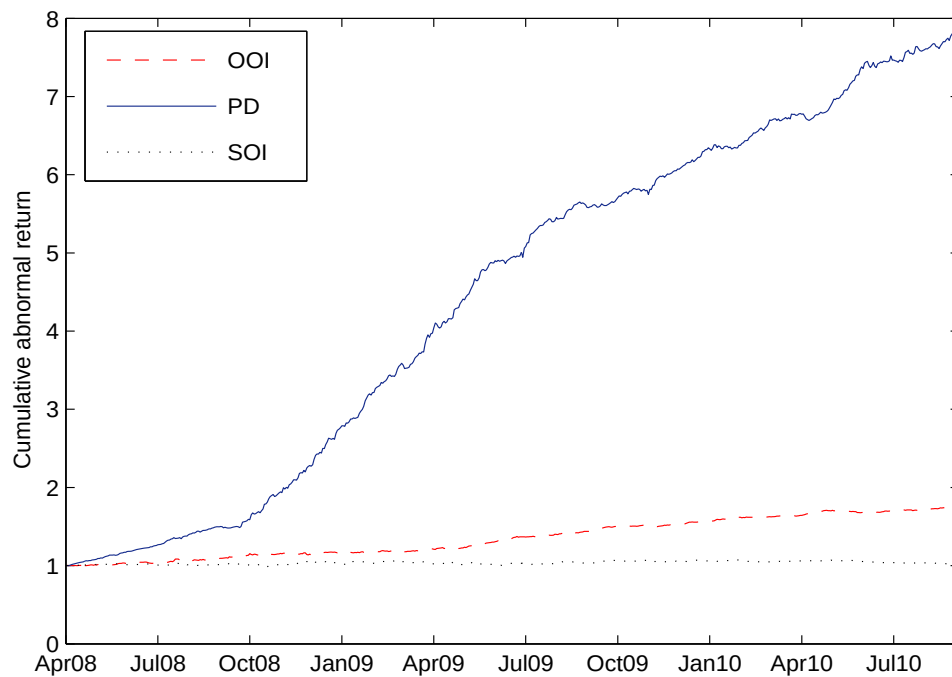
**Figure 2**  
**Time series of t-statistics of regression coefficients.**



**Figure 3**

**The predictive ability of option order imbalance and price discrepancy for future stock returns.**

Plotted are the estimated coefficients and 95% confidence intervals of option order imbalance and price discrepancy in equation 5 with dependent variables being one to twenty days ahead risk adjusted stock returns. The solid line represents the estimated coefficients and the two dash lines represent the upper and lower bounds of the confidence intervals.



**Figure 4**

**Cumulative abnormal returns on portfolios based on order imbalance and price discrepancy.**

Plotted are the cumulative abnormal returns on long-short portfolios based on option order imbalance, stock order imbalance, and price discrepancy during our sample period. All portfolios are formed at market close and held for one day.